Learner-Centered Automatic Readability Assessment: Leveraging Hybrid Models to Consider L1

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Abstract

Automatic Readability Assessment (ARA) traditionally focuses on linguistic dimensions of texts, whether lexical or grammatical. However, text difficulty is not solely a property of the text itself; it also depends on the characteristics of the learner. We propose a learner-centered approach to readability assessment that incorporates a learner's first language (L1). Our method leverages large language models (LLMs) to translate texts into different languages and computes normalized Levenshtein Distance (LDN) between the original and translated texts. The approach is motivated by the cognate facilitation effect (CFE) stating that cognates facilitate lexical processing for language learners. LDN serves as an approximation of cognates and cross-linguistic lexical similarity. We discuss how these language-specific distance measures can be used as features for learner-centered readability assessment.

1 Introduction

Automatic Readability Assessment (ARA) evaluates reading difficulty of texts to support teachers or learners to find appropriate reading material. Selecting texts aligned with learners' reading levels is an important condition for successful language learning (Sangers et al., 2025). Reading can contribute to the development of linguistic knowledge, particularly vocabulary, but acquisition through reading depends on learners noticing and cognitively processing of unfamiliar lexicon (Tack, 2021). Vocabulary knowledge plays a central role in reading comprehension, but learners' judgments of word difficulty can vary according to their existing linguistic knowledge and experience (Tack, 2021).

To automatically measure readability, different approaches have been proposed and successfully implemented into ARA systems. Readability formulas have a long tradition in ARA research, as they are easy to compute through leveraging surface-level features, such as the number of syllables per word and words per sentence (Collins-Thompson, 2014). Other approaches have used measures of morphology, syntax, lexicon, discourse, or combinations of these (Akef et al., 2024; Löfflad et al., 2025). Models range from highly interpretable statistical to deep learning models (Weiss et al., 2021; Vázquez-Ingelmo et al., 2026; Liu et al., 2025).

While many of these approaches reach good performance on annotated datasets, they do not acknowledge learners' linguistic background. The reading difficulty of a text cannot be captured by linguistic dimensions alone but always comes from an interaction of text and learner. The learner-side has understandably gained less attention, as it is difficult to capture in computer-based standalone systems, which often is the case for ARA systems. In contrast, Intelligent Tutoring Systems (ITS) often have a learner model that contains information about the learners, such as topic knowledge, language proficiency, and L1 (Colling et al., 2023). ARA systems are unable to consider all these factors without a learner model. Building learner models is, however, laborious and requires the learner to interact with the system over a longer period, which is often not feasible.

We here thus ask the question:

RQ: How can linguistic background of learners be taken into account for learner-centered ARA systems?

To tackle this question, we propose to leverage crosslinguistic transfer for learner-centered ARA. Crosslinguistic influence describes how a learner's previously acquired language(s) can facilitate or inhibit the processing, acquisition, and use of an additional language (Odlin, 1989; Jarvis and Pavlenko, 2008). Among the different forms of crosslinguistic influence, lexical transfer concerns similarities

between words across languages. It is motivated by the Cognate Facilitation Effect (CFE), which states that words sharing both form and meaning across languages (cognates) are processed more efficiently by multilingual speakers (Jarvis and Pavlenko, 2008; Carroll, 1992).

In computational systems, cognates can be approximated using normalized Levenshtein Distance (LDN, Beijering et al. 2008; Schepens et al. 2012). Since computing LDN requires corresponding lexical items across languages, we first translate the source text using an LLM and then compute LDN on aligned word pairs. We thus propose a hybrid model that combines the multilingual capabilities of LLMs with traditional, interpretable algorithms for learner-aware readability assessment.

This division is deliberate. Although an LLM could, in principle, be prompted to estimate cognacy directly, this would be an under-researched black-box approach. This is especially relevant in multilingual applications, because the performance of multilingual models depends strongly on the availability of language-specific training data and is therefore not necessarily comparable across languages, and usually biased towards English (Chen and Li, 2026). We instead restrict the LLM to translation and lexical alignment and derive similarity using a deterministic and reproducible measure. LDN is a strongly theoretically motivated operationalization that has repeatedly been validated against expert cognacy judgments and other measures of crosslinguistic similarity (Beijering et al., 2008; Schepens et al., 2012, 2020). The hybrid design thus combines the multilingual coverage of LLMs with an explicit similarity measure whose interpretation remain stable across learner groups.

2 Background

Crosslinguistic transfer refers to the influence that a learner's previously acquired language(s) has on the acquisition, processing, and use of a second or third language (L2 and L3, respectively) (Engemann and Radetzky, 2024). Within generative approaches, crosslinguistic transfer can be operationalized through variation in features of language systems across linguistic domains (Alexopoulou and Gil, 2026). These dimensions include syntax, morphology, phonology, and the lexicon, all of which can contribute to crosslinguistic influence. From a usage-based perspective, the term 'crosslinguistic interaction' is favoured, on the assumption of a unified linguistic system (Jessner et al., 2016).

To investigate this influence, prior work has employed measures of linguistic distance, capturing similarities and differences between languages across these dimensions. Among the various domains, lexical similarity has been shown to play a particularly important role in L2 acquisition and processing (Schepens et al., 2020; Shatz et al., 2024; Castle et al., 2026). Lexical transfer concerns the interaction between vocabulary in a learner's L1 and the target language. Jarvis and Pavlenko (2008) further argue that cross-linguistic influence affects lexical representation, lexical accessibility, and lexical activation. Cognates facilitate lexical access because representations from different languages partially overlap, allowing activation to spread across languages during word recognition.

The CFE provides the psycholinguistic motivation for modeling lexical similarity in learner-aware readability assessment. Cognates, i.e. words in two languages sharing similarities in both form and meaning, can facilitate processing, acquisition, and production in the L2 (Engemann and Radetzky, 2024; Dressler et al., 2011). This is typically described as positive transfer. In contrast, negative transfer occurs when L1 properties interfere with L2 acquisition. For example, while a cognate such as *information* may facilitate vocabulary acquisition for a Spanish learner of English, it may simultaneously transfer incorrect grammatical features, such as countability, thereby inhibiting full acquisition (Alexopoulou and Gil, 2026).

A key challenge in studying lexical transfer computationally is that cognacy is not directly observable from surface forms. To address this, prior work has proposed using normalized Levenshtein Distance (LDN) as an approximation of lexical similarity (Beijering et al., 2008; Schepens et al., 2012). LDN measures the minimum number of character insertions, deletions, and substitutions required to transform one string into another, normalized by the length of the longer string. For instance, the English–German phonological pair /ni/ (knee) and /kni/ (Knie) yields an LDN of 0.33, reflecting one edit operation over a maximum string length of three (Shatz, 2021).

Although LDN does not directly identify cognates, it captures form-based similarities that are relevant for lexical processing and can therefore serve as a proxy for crosslinguistic lexical similarity. Previous studies have shown that LDN correlates strongly with expert cognacy judgments (Bei-

jering et al., 2008; Schepens et al., 2012) as well as with measures based on morphological features (Schepens et al., 2020). Moreover, research comparing phonological and orthographic overlap has demonstrated substantial correlations between these representations, suggesting that orthographic similarity can serve as a reasonable proxy for phonological similarity in languages sharing the same script (Shatz et al., 2023).

Given these findings, LDN provides a practical and scalable way to approximate lexical transfer effects in computational systems. Incorporating such measures into ARA offers a promising direction to capture learner-centered variation without requiring explicit learner models, thereby enabling more learner-aware readability predictions.

3 System Description

Our system is designed to support learner-centered ARA by calculating lexical similarity between a source text and the learner's first language. Users upload a German text through a web interface. The system first translates the text into the target language using Gemini 2.5 Flash. In addition to the translation, the language model is prompted to generate an alignment between the original and translated text, mapping corresponding words or short multi-word expressions. Defining multi-word expressions is a wide research field, and multi-word expressions can be based on different characteristics such as formulaic expressions, irregularities or fixedness (Schmid, 2026). Multi-word units are grouped whenever this alignment better captures the semantic correspondence than individual word alignments. For the present application, we adopt a pragmatic approach and prompt the LLM to group words into translation units where appropriate. For example, the German *um zu testen* translates to English *to test* or French *pour tester*, such that treating *zu testen* as a single unit yields a more meaningful alignment.

Based on these alignments, we compute the LDN for each word pair. LDN is calculated as the minimum number of character insertions, deletions, and substitutions required to transform one word into the other, divided by the length of the longer word. This normalization accounts for differences in word length and yields values between 0 and 1, where lower values indicate greater lexical similarity.

The computed LDN scores are then visualized in the web interface, see Figure 1. Words with an LDN above 0.9 are explicitly listed as potentially challenging lexical items. This enables teachers to inspect these words individually and, if desired, replace them with more accessible alternatives for learners with a particular first-language background or to offer additional scaffolding through Language-sensitive subject teaching (Sprachsensiblen Fachunterricht, Leisen 2024).

The resulting system combines the translation and alignment capabilities of LLMs with a transparent, interpretable similarity measure, providing a practical approach to incorporate learner-specific lexical information into ARA without requiring explicit learner models.

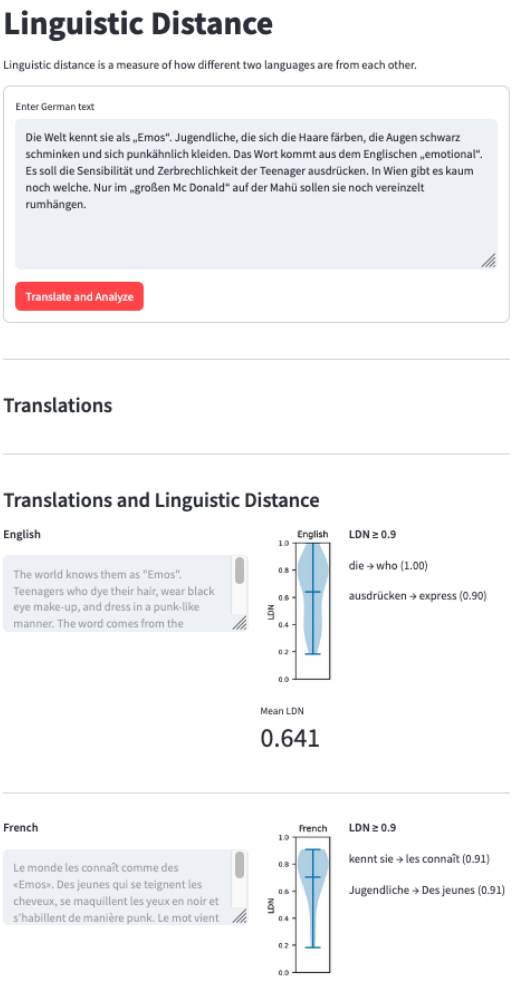


Figure 1: Web interface of German text upload, LLM-based translation and LDN visualisation.

4 Future Validation

The current work does not yet include an empirical evaluation of the proposed approach, as it is currently a work in progress. Future work will investigate the usefulness of lexical similarity fea-

tures for readability assessment through several complementary validation strategies.

The proposed lexical similarity measure can be integrated into existing ARA models as an additional feature alongside established linguistic complexity measures. Its contribution can then be evaluated on annotated readability datasets using regression or other supervised learning approaches to quantify its predictive value beyond traditional readability features.

The lexical similarity scores could be compared against external lexical resources. For example, words identified as highly dissimilar between languages could be analyzed with respect to their CEFR vocabulary levels or corpus-based frequency measures. If high-LDN words systematically correspond to lower-frequency vocabulary or higher CEFR levels, this would provide further evidence that the proposed measure captures aspects of lexical difficulty relevant for language learners.

Moreover, a feasibility study with German as a Second Language (GSL) learners and teachers can be performed to assess whether the LDN score captures text understanding and perceived text difficulty.

Together, these evaluation strategies would provide evidence for both the predictive value of the proposed feature in computational readability assessment and its usefulness in educational settings. In particular, they would allow us to investigate whether learner-centered lexical similarity measures not only improve readability prediction, but also support teachers in selecting and adapting reading materials while reflecting learners' perceived text difficulty and comprehension.

5 Discussion

The proposed approach is a first step towards incorporating learner characteristics into ARA without relying on explicit learner models. Rather than predicting readability solely from linguistic properties of the text, our approach considers the interaction between the text and the learner by approximating lexical transfer through crosslinguistic similarity. While simple, this perspective aligns more closely with psycholinguistic theories of reading, which view reading difficulty as emerging from both the text and the reader (Amiran and Jones, 1982).

For readability assessment of written texts, we argue that orthographic similarity is an appropriate representation of lexical transfer. Unlike spoken language applications, ARA is concerned with reading, where orthographic word forms directly influence word recognition and processing. Although previous work has often employed phonological similarity, orthographic similarity is strongly correlated with phonological similarity for languages sharing the same script and can therefore serve as a practical approximation while being easier to compute (Shatz et al., 2023).

An important advantage of the proposed pipeline is its flexibility. Since the translation and alignment are generated automatically, the same source text can be processed for different first languages simply by selecting another target language. This enables dynamic, learner-specific analyses without requiring separate readability models or manually curated lexical resources for each language.

This flexibility is particularly valuable in educational settings, where classrooms often consist of heterogeneous learner groups with diverse linguistic backgrounds. Teachers could inspect lexical similarity for multiple L1s and identify vocabulary that may pose different levels of difficulty for different groups of learners. This is particularly relevant in integration classes, where adult learners from various linguistic backgrounds often learn together (BAMF, 2024). By allowing text selection and adaptation that account for learners' linguistic backgrounds, learner-centered ARA has the potential to improve the accessibility of reading materials and support more individualized language learning.

The educational relevance of this flexibility extends beyond language-learning contexts. Looking at PISA data, Borgonovi and Ferrara (2020) found that greater linguistic distance between students' first language and the language of instruction was negatively associated with achievement not only in reading, but also in mathematics and science. This suggests that linguistic distance may affect learners' access to instructional texts across subjects. Accordingly, our proposed pipeline could support teachers in content subjects by helping them to identify words whose readability is likely to differ across students' linguistic backgrounds. This information could inform vocabulary scaffolding, explanations, or the adaptation of materials, particularly for learners whose first language is relatively distant from the language of instruction. Similar results have been found in German-speaking contexts (Jaekel et al., 2024; Berthelé et al., 2022).

An additional advantage of the proposed pipeline is its hybrid design. The LLM is not employed to

predict readability directly, but instead performs the tasks of translation and lexical alignment, which would otherwise require dedicated multilingual resources or word alignment systems. LLMs have been shown to achieve high results in translation tasks, even for ambiguous texts containing polysemic words (Iyer et al., 2023). The resulting alignments are then processed by an interpretable LDN algorithm to derive learner-centered lexical similarity scores. This combination leverages the multilingual capabilities of LLMs while maintaining the transparency, reproducibility, and extensibility of traditional NLP approaches. Consequently, the proposed feature can be seamlessly integrated into existing readability assessment pipelines rather than replacing them with an end-to-end black-box model.

The proposed lexical similarity measure is not intended to replace established readability features, but rather to complement them. It can be incorporated into existing feature extraction pipelines along with predictors such as word frequency, lexical sophistication, syntactic complexity, and discourse features (Weiss et al., 2021; Löfflad et al., 2025; Vázquez-Ingelmo et al., 2026). In this sense, the approach provides a lightweight mechanism for introducing learner-centered information into readability assessment without the substantial effort required to build and maintain learner models. This is of particular interest given that current psycholinguistic studies have shown that the CFE may interact with other factors such as frequency, word type and student proficiency.

Beyond readability prediction, lexical similarity may also provide an additional perspective on lexical difficulty. Existing approaches frequently rely on corpus-based measures such as word frequency, age of acquisition, or CEFR vocabulary levels (Weiss, 2024). While these measures capture general difficulty, they do not account for differences between learner groups. In contrast, lexical similarity explicitly models accessibility relative to a learner's linguistic background and may therefore capture aspects of lexical difficulty that traditional measures overlook, particularly in multilingual learning contexts. Recent work further suggests that cognate facilitation is modulated by learner proficiency and word frequency (Gastmann et al., 2025). This supports our view that lexical similarity should complement rather than replace established readability features such as word frequency and lexical sophistication.

Limitations

This work is currently a work in progress and a proposition on how to integrate learner-centered features into ARA systems, and validations are planned. Moreover, this approach works well for languages using the same alphabetic system. To compare languages using different alphabetic systems, a mapping will be required.

Lay Summary

An important part of learning a new language is reading texts in that language. Teachers often need a lot of time to find texts that match their students' interests and reading skills. Computer programs can help teachers choose suitable texts. For example, they can count the number of words in a text or in each sentence. However, these programs usually do not consider the learner's first language, also called L1. A learner's first language can affect which texts and words they find easy or difficult to read.

Some words look similar and have the same meaning in two languages. These words are called cognates. For example, the German word Information may be easier for a French-speaking learner to understand because it is similar to the French word information. Cognates can make words easier to understand, this effect is called the cognate facilitation effect.

We propose a new way to include the learner's first language when measuring how easy or difficult a text is to read. We use large language models to translate a German text into another language, such as French. We then match each German word with its translation. Next, we compare the letters in the two words, as similar words should be easier for the learner to understand. We measure this similarity with a method called normalized Levenshtein distance. This method counts how many letters must change to turn one word into another. Previous research has shown that this method can help to identify cognates.

The system has not yet been tested with learners or teachers. Future studies will examine whether this approach can help measure the readability of German texts for learners with different first languages. In conclusion, we combine large language models with traditional computer-based language methods. This allows us to include the cognate facilitation effect when measuring readability. We use normalized Levenshtein distance to identify

possible cognates.

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